

OncoEvidence: Mechanism-Guided AI Evidence Triage for Cancer Drug Repurposing

A knowledge-graph and language-model pipeline that prioritizes repurposing candidates by mechanistic plausibility and literature evidence

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The problem. Drug repurposing looks for new uses of existing drugs, for example asking whether an approved drug could treat a cancer it was not designed for. Computational methods can rank thousands of drug-disease pairs, but a high score is not enough: a researcher still needs to know *why* a drug might work (its mechanism) and whether the literature supports it. This project builds a pipeline that, for cancer repurposing, ranks candidates, extracts the biological mechanism path that would explain each one, and checks that mechanism against published evidence.

Proposed study

I will build an oncology-focused evidence-triage pipeline on the PrimeKG biomedical knowledge graph. It will (1) rank drug-cancer candidates and compare a graph model against a strong tabular baseline to test honestly when the graph helps, (2) extract multi-hop *drug* $\rightarrow$ *target protein* $\rightarrow$ *cancer gene* $\rightarrow$ *cancer* mechanism paths for the top candidates, and (3) use a language model to read the literature and grade whether each mechanism is supported. I will evaluate the system against known indications and a curated mechanism database. The deliverable is a ranked, evidence-backed shortlist of repurposing candidates with transparent mechanistic rationales.

Why this project. I am drawn to drug repurposing because it can bring existing, already-safe drugs to new cancer patients faster than designing a drug from scratch. What interests me technically is the gap between a prediction and an explanation: a model can output a score, but a clinician or scientist needs a reason. I want to build a system that produces the reason and then checks it against the evidence.

1 Background and Significance

Cancer drug repurposing matters because developing a new drug takes many years and large cost, while thousands of approved drugs already have known safety profiles. If one of them acts on a protein or pathway that also drives a cancer, it may be a fast route to a new treatment. Biomedical knowledge graphs such as PrimeKG collect millions of relationships among drugs, proteins, genes, pathways, and diseases, so the information needed to reason about such mechanisms is, in principle, already encoded as paths through the graph.

The challenge is trust. A model that only outputs a drug-disease score is hard to act on, because the reviewer cannot see the reasoning and cannot tell a real mechanism from a coincidence. Recent work has shown that graph models can predict repurposing links and even suggest paths, and that language models can reason over biomedical text. What is still under-explored is an honest, evaluated combination: a system that produces a mechanism and then verifies it against the literature, and that reports clearly when the graph is actually necessary.

2 Central Question and Specific Aims

Core question. For cancer drug repurposing, can a biomedical knowledge graph combined with a literature-grounded language-model agent prioritize candidates by both predicted therapeutic relevance and mechanistic plausibility, and when does the graph add value over a strong tabular baseline?

Specific Aims

Aim 1 (Candidate generation). Rank drug-cancer pairs with PrimeKG models (a heterogeneous graph neural network and a knowledge-graph embedding) and a tuned XGBoost baseline on the same features, and report honestly where the graph helps.

Aim 2 (Mechanism extraction). For top candidates, extract multi-hop $drug \rightarrow target \rightarrow pathway/cancer-gene \rightarrow cancer$ paths, prioritizing real mechanism relations over phenotype co-incidence.

Aim 3 (Evidence verification). Retrieve literature and have a language model grade each path as supported, weak, contradicted, or unknown, with a quoted piece of evidence.

Aim 4 (Evaluation). Test the system against known indications and a curated mechanism resource (DrugMechDB), measuring whether mechanism paths separate true indications from random pairs and whether verified mechanisms agree with curated ones.

3 Datasets

Resource	Content	Role
PrimeKG	biomedical knowledge graph: drugs, proteins, genes, pathways, diseases ($\sim 129k$ nodes, $\sim 8M$ edges)	Candidate ranking + mechanism paths
Europe PMC	open biomedical literature API (titles, abstracts, open-access full text)	Evidence retrieval for verification
DrugMechDB	expert-curated drug mechanism-of-action paths	Ground truth for mechanism evaluation

Table 1: All resources are public. PrimeKG supplies the graph; Europe PMC supplies the evidence; DrugMechDB supplies curated mechanisms to evaluate against.

4 Proposed Methodology

Knowledge graph and candidate ranking. I will represent PrimeKG as a heterogeneous graph and frame repurposing as link prediction on the $drug \rightarrow indication \rightarrow disease$ edges: hide some known treatment links and score drug-disease pairs that are not currently connected. I will give the graph model and the tabular baseline the *same* node features (text embeddings of names and descriptions), so any difference measures the value of graph structure rather than the features.

Mechanism extraction. For top-ranked candidates I will search the graph for short, specific paths connecting a drug to a cancer through its target protein, protein interactions, and shared pathways, while down-weighting generic hub proteins (such as common drug carriers) so the paths stay mechanistic.

Evidence verification. For each mechanism path I will retrieve relevant abstracts from Europe

PMC and ask a language model to decide whether the text supports that specific mechanism, requiring it to quote the supporting sentence and to treat a mere statistical association as weak rather than supporting evidence.

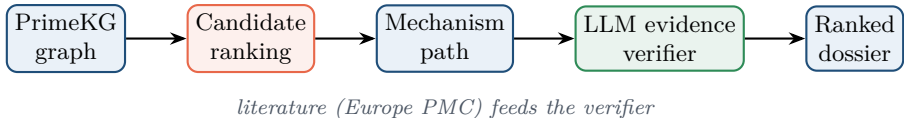


Figure 1: Planned pipeline: rank candidates on the graph, extract a mechanism path for each, verify the path against the literature, and output an evidence-backed shortlist.

5 Planned Evaluation

I will not rely on a single accuracy number. The plan is to test the system from several angles (Table 2), with an emphasis on whether the mechanisms are real, not just whether the scores are high.

Question	How I will test it
Is the graph necessary for ranking?	Graph model vs tuned XGBoost on the same features, under leakage-safe splits.
Do mechanism paths mean anything?	Do true indications carry stronger mechanism paths than random drug-cancer pairs?
Are the mechanisms real?	Agreement of extracted paths with curated DrugMechDB mechanisms.
Is the evidence verification trustworthy?	Among “supported” calls, how many are confirmed by curated mechanisms.

Table 2: Planned evaluation. The goal is an honest picture of where the system works, including reporting any place where a simple baseline is already enough.

6 Relationship to Prior Work

This project does not claim to invent knowledge-graph or language-model drug repurposing; close prior work exists and I treat it as reference. TxGNN (Nature Medicine, 2024) is a graph model for zero-shot repurposing with multi-hop explanations; KGML-xDTD predicts treatment links with mechanism paths; DrugKLM combines knowledge graphs with language-model reasoning; and Decagon showed graph models excel on relational drug-pair tasks. My contribution is narrower and honest: an oncology-specific, literature-grounded evidence-triage pipeline that tests when the graph is actually necessary and evaluates whether the proposed mechanisms are supported by retrieved literature and curated databases. I combine and evaluate; I do not claim to be first.

7 Eight-Week Timeline

Weeks	Milestones
1–2	Build the PrimeKG pipeline; set up leakage-safe splits and shared features.
3–4	Candidate ranking (graph model, knowledge-graph embedding) and the tuned XGBoost baseline.
5	Mechanism-path extraction with hub down-weighting; sanity-check on known drugs.
6	Literature retrieval and the language-model verifier (supported/weak/contradicted/unknown).
7	Evaluation: separation test, DrugMechDB agreement, verifier precision.
8	Deliverable shortlist, figures, and a written report.

Table 3: Schedule for the eight-week SRI cohort.

8 Reproducibility and Ethics

I will use open tools (PyTorch Geometric, scikit-learn, XGBoost) and public data (PrimeKG, Europe PMC, DrugMechDB), and will release code, configurations, and random seeds. All predictions are research-only and hypothesis-generating; they are not medical advice and would require experimental and clinical validation before any use.

References

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